INDROVE: A new era of India's Pipeline health.

Methodology:

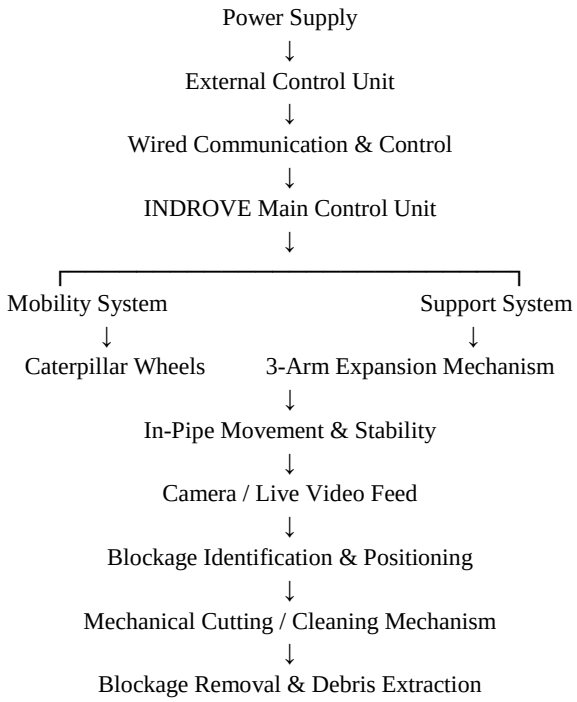
INDROVE is a modular robotic rover designed to operate inside horizontal cylindrical pipelines for inspection and blockage removal without human entry. The project focuses on developing a compact, mechanically stable and remotely controlled robotic system for safer and more efficient pipeline maintenance.

The rover uses a caterpillar-type wheel mechanism for movement and traction inside the pipeline. A three-arm expandable support mechanism provides multi-point contact with the inner pipe wall, helping the rover maintain stability and adapt to variations in pipeline diameter. The arms can be expanded or retracted according to the pipe size.

An onboard camera provides live visual feedback to an external operator, enabling inspection and identification of blockages. After reaching the affected location, the Operation configuration can perform mechanical blockage removal using a suitable cutting/cleaning mechanism, with provision for debris extraction. The system is operated through an external wired controller and power supply.

The present project targets pipelines in the approximate 200–300 mm diameter range, with the design being developed towards a practical and modular field-deployable system.

- ### Proposed Block Diagram:



Proposed Outcome:

The project aims to develop a functional prototype of ***INDROVE*** capable of controlled movement inside horizontal cylindrical pipelines, maintaining stability across different pipe diameters, providing live visual inspection and performing targeted blockage-removal operations.

The proposed system can reduce the need for human entry into confined pipelines and enable direct, location-specific maintenance rather than relying only on surface-level removal through manholes. The project will demonstrate the feasibility of robotic intervention for safer, more efficient and sustainable pipeline maintenance, with scope for future integration of additional sensors and automated inspection capabilities.